

MARIN LAUBER

Born 28th August 1994 ♦ Swiss ♦ Currently in the Netherlands

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INTRODUCTION

I am a passionate researcher in computational mechanics from Switzerland. My expertise lies in the computational fluid-structure interaction of highly deformable and lightweight membranes and shells.

Throughout my academic journey, I've gained experience in developing high-fidelity computational methods to tackle problem in maritime engineering, bio-locomotion and cardiovascular biomechanics. An extensive part of my expertise is in the development of scientific software (finite-volume, finite-element, isogeometric analysis) to tackle these challenging engineering problems. I'm currently one of the main developer of a Julia open source flow solver and a Julia shell finite-element library.

I have experience with data-driven approaches for data analysis and high-performance computing with graphics cards for accelerating physics-based solvers. Finally, I am an avid watersport enthusiast, and I have an interest in sailing performance prediction.

EXPERIENCE

Post-doctoral researcher at TU Delft

September 2024 - Present

Technische Universiteit Delft

Delft, Netherlands

- Post-doctoral researcher within the Biomechanical Engineering research group, working on numerical modelling of prosthetic human heart.
Head of the lab: Dr. Mathias Peirlinck

Post-doctoral researcher at TU Delft

February 2023 - September 2024

Technische Universiteit Delft

Delft, Netherlands

- Post-doctoral researcher within the Ship Hydromechanics research group, working on fluid-structure interaction of thin membranes and shells applied to sail-assisted propulsion.
Head of the lab: Prof. Gabriel D. Weymouth

Doctoral researcher at the University of Southampton

Septembre 2018 - February 2023

Univeristy of Southampton

Southampton, UK

- Doctoral researcher within the Maritime Engineering research group, working of fluid-structure interaction of thin membranes and shells.
Head of the lab: Prof. Gabriel D. Weymouth

Visiting PhD Student at the Unsteady Flow Diagnostic Lab September 2021 - May 2022
Ecole Polytechnique Fédérale de Lausanne *Lausanne, Switzerland*

· Visit organized for the last year of my Ph.D., allowed me to work with experimental researchers on validating the numerical model developed during my Ph.D. on a heaving flexible swimmer undergoing complex accelerations.

Head of the lab: Prof. Karen Mulleners

Development of a Velocity Prediction Program March 2020 - Present
Side project

· Development of an open-source 3 degrees of freedom velocity prediction program (non-linear multi-variate constraint optimization problem) in Python. See: github.com/marinlauber/Python-VPP

EDUCATION

PhD in Next Generation Computational Modelling *September 2018 - February 2023*
University of Southampton *Southampton, UK*

Thesis: Computational Fluid-Structure Interaction of Membranes and Shells with Application to Bat Flight

Taught Year: Lectures in Simulation and Modelling, Numerical Methods, Statistical Analysis, Computational Methods, Advanced Finite-Element Analysis, Partial Differential Equations, Integral Transform Methods.

Experience Gained: Development of a finite-volume (LES) Cartesian grid fluid flow solver (Fortran) to simulate thin, flexible sheets/membranes in an unsteady flow. Coupling of the finite-volume code to a finite-element solver (CalculiX) through the preCICE library to simulate weak and strongly coupled fluid-structure interaction problems.

MSc in Maritime Computational Fluid Dynamics *September 2017 - September 2018*
University of Southampton *Southampton, UK*

Thesis: Acquisition of Manoeuvring Characteristics of Ships using RANS CFD

Overall grade: *First Class*

Lectures in Turbulence, Computational Fluid Dynamics, Aeroelasticity, Ship Resistance and Propulsion, Finite-Element Analysis, Hydrodynamics, and Fluid-Structures Interactions.

Experience Gained: Performing static drift and planar motion mechanism simulation of ship manoeuvring using a commercial CFD package (Star-CCM) using overset meshes.

BEng in Yacht & Powercraft Design *September 2014 - September 2017*
Southampton Solent University *Southampton, UK*

Final Year Project: Preliminary Design of a Mini 6.50 with a Foil CFD Investigation

Award: The 2017 RINA - BAE Systems Student Naval Architect Award - Final Year Project

Overall grade: *First Class with Honours*

Certificat de Maturité Gymansiale

Gymnase de Morges

Option: Physics & Applied Mathematics, Economy

August 2010 - September 2014

Morges, Switzerland

Overall grade: 4.1/6

LIST OF PUBLICATIONS

First-author Paper (peer-reviewed)

- Lauber M., Arfaee M., Peirlinck, M. (2026) *The mechanics of the Less In More Out artificial heart: Modeling fabric-based soft robotic devices*, Journal of the Mechanics and Physics of Solids, 10.1016/j.jmps.2026.106565
- Lauber M., Weymouth G.D., Limbert G. (2024) *Immersed boundary fluid structure interaction of membranes and shells*, Journal of Physics: Conference Series, 10.1088/1742-6596/2647/5/052002
- Lauber M., Weymouth G.D., Limbert G. (2023) *Rapid flapping and fiber-reinforced membrane wings are key to high-performance bat flight*, Journal of the Royal Society Interface, 10.1098/rsif.2023.0466
- Lauber M., Weymouth G.D., Limbert G., (2022) *Immersed boundary simulations of flows driven by moving thin membranes*, Journal of Computational Physics, 10.1016/j.jcp.2022.111076

Co-Author Papers (peer-reviewed)

- Weymouth G.D., Lauber M. (2026) Stability of Kirigami parachutes in effectively infinite numerical domains . arXiv preprint arXiv:2606.17661
- Weymouth G.D., Lauber M. (2025) *Far-field boundary conditions for Cartesian grid simulations*, ArXiv, draft,
- Byrne C., Dickson T., Lauber M., Cairoli C., Weymouth G.D. (2022) *Using Machine Learning to Model Yacht Performance*, Journal of Sailing Technology, 10.5957/jst/2022.7.5.104

Conferences

- Lauber M., Arfaee M., Peirlinck M. (2026) *Modeling babric-based total artificial hearts*, 17th World Congress on Computational Mechanics and 10th European Congress on Computational Methods in Applied Sciences and Engineering, Munich, Germany.
- Lauber M., Li J., Verhelst H.M (2024) *Gismo & WaterLily adapters for the preCICE coupling library*, 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics, Vancouver, Canada.
- Buchner, A.J., Lauber M., de Boer D.; Cribellier A., Muijres F. (2024) *The low amplitude wing beating of mosquitoes and its effect on wing-wake interaction*, 1st European Fluid Dynamics Conference, Aachen, Germany.
- Lauber M., Weymouth G.D. (2024) *Fast Fluid-Structure Interaction in Minimal Domains with Potential-Flow Boundary Conditions*, Third Direct In-person Colloquium on Vortex Dominated Flows (Dis-CoVor), Delft, The Netherlands.
- Lauber M., Weymouth G.D (2023) *Strongly-coupled fluid-structure interaction of lightweight membranes and shells*, American Physical Society - Division of Fluid Dynamics Annual Meeting 2024, Washington-DC, USA.

- Weymouth G.D, Lauber M. (2023) *High-performance bat flight simulations using parametric kinematic and material models*, Bio-inspired aerial and aquatic locomotion workshop 2023, Les Houches School of Physics, France.
- Lauber M., Weymouth G.D., Limbert G., (2023) *Strouhal and Membrane Elasticity effects on bat flight*, Second Direct In-person Colloquium on Vortex Dominated Flows (DisCoVor), Breckenridge, USA.
- Lauber M., Weymouth G.D., Limbert G., (2022) *Development and application of an immersed boundary fluid-membrane interaction solver*, First Direct In-person Colloquium on Vortex Dominated Flows (DisCoVor), Villars-sur-Ollon, Switzerland.
- Lauber M., *Flexible sheets in Turbulent flow*, UK Fluid 2021, Southampton, UK
- Lauber M., Weymouth G.D. (2020) *Improving Pressure Simulations Driven by Immersed Dynamic Surfaces*. 73rd Annual Meeting of the APS Division of Fluid Dynamics, Chicago, USA.

REFERENCES

Prof. Mathias Peirlinck

Postdoc PI

contact: M.Peirlinck@tudelft.nl

- Professor of Cardiovascular Biomechanics at the Delft University of Technology

Prof. Gabriel D. Weymouth

PhD advisor

contact: g.d.weymouth@tudelft.nl

- Professor of Ship Hydromechanics at the Delft University of Technology

Prof. Georges Limbert

PhD advisor

contact: g.limbert@soton.ac.uk

- Professor of Computational Mechanics at the University of Southampton

AWARDS

Best Early Career Paper Award

XII International Conference on Structural Dynamics

2023

- For the paper: *Immersed boundary fluid-structure interaction of membranes and shells*

Best Career Paper Award

XII International Conference on Structural Dynamics

2023

- For the paper: *Immersed boundary fluid-structure interaction of membranes and shells*

RINA - BAE Systems Student Naval Architect Award - Final Year Project

Royal Institution of Naval Architects

2017

- Awarded for the project: *Preliminary Design of a Mini 6.50 with a Foil CFD Investigation*

SKILLS

Languages	French (native), English (proficient user), German (independent user)
Programming	Julia (MPI, CUDA), Fortran (OpenMP, MPI), Python, <i>C</i> , Matlab
Version control	Git, GitHub (https://github.com/marinlauber)
Modeling	Autocad, Rhinoceros 5, Maxsurf (modeler, stability, seakeeping, structure), Solidworks
CFD	Star-CCM, Ansys Fluent, Ansys CFX, OpenFOAM
FEA	CalculiX, ABAQUS, Ansys APDL
Experimental	Towing Tank (resistance, seakeeping)
Other	HullScant (ISO 12215), WinDesign (VPP)

ADDITIONAL INFORMATION

Interests	Sailing, kitesurfing, wingfoil, skiing
Driving Licence	A1, B , Sailing & motor yachts (Swiss)